

METRO TICKET GENERATING SYSTEM IN SERVICENOW

FINAL PROJECT DOCUMENT

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Project Name	Metro Ticket Generating System in ServiceNow
Platform	ServiceNow

1. INTRODUCTION

1.1 Project Overview

The Metro Ticket Generating System is a ServiceNow-based digital ticketing solution designed to simplify metro ticket booking and generation. The system provides a centralized Service Catalog interface where passengers can enter journey details, select source and destination stations, specify the number and type of tickets, and complete the ticket-generation process. The system uses ServiceNow automation to reduce manual effort and provide a faster, more convenient ticketing experience.

1.2 Purpose

The purpose of this project is to digitize the metro ticket-booking process by providing automated fare calculation, ticket generation, QR-code-based ticket validation, payment-status tracking, ticket expiry management, and smart-card recharge support. The solution aims to reduce waiting time, minimize manual errors, and maintain ticket information in a structured ServiceNow environment.

2. IDEATION PHASE

2.1 Problem Statement

Traditional metro ticket booking can involve queues, manual ticket handling, fare-calculation errors, and difficulty in tracking ticket information. Passengers may spend additional time at ticket counters, while manual processes can increase operational effort. A digital and automated ticketing solution is therefore required to make metro ticket generation faster, reliable, and accessible.

2.2 Empathy Map Canvas

The empathy analysis focuses on the needs and difficulties of metro passengers:

- User Says: "I want to book my metro ticket quickly without waiting in a long queue."
- User Thinks: "I need a simple way to know my fare, payment status, and ticket validity."
- User Does: Selects source and destination, enters ticket details, completes payment, and uses the generated ticket/QR code.
- User Feels: Frustrated when ticket booking takes too long or when fare and ticket details are unclear.

2.3 Brainstorming

The ideation process identified the following core functional requirements:

- Unified Ticket Booking Form: A Service Catalog item captures journey and passenger details in one place.
- Automated Fare Calculation: Fare is calculated according to the selected journey and number of tickets.
- Automated Ticket Generation: A ticket record is generated after the booking process.
- QR Code Validation: The generated ticket can be represented and confirmed using QR-code functionality.
- Digital Payment and Status Tracking: Payment status and booking information are maintained in ServiceNow.
- Smart Card Recharge: The system can support recharge-related ticketing operations.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The passenger journey is organized into four major stages:

- Discovery: The passenger opens the ServiceNow ticket-booking catalog item.
- Configuration: The passenger selects source station, destination station, ticket type, number of tickets, and other required details.
- Booking and Payment: The system calculates the fare and processes the selected payment option/status.
- Ticket Fulfillment: The system generates the metro ticket, stores booking information, and provides QR-code-based validation and expiry details.

3.2 Solution Requirements

Functional Requirements (FR): The system should capture station and passenger information, calculate fare automatically, generate metro tickets, maintain payment status, generate/validate QR-code information, store booking dates and expiry details, and support smart-card recharge operations.

Non-Functional Requirements (NFR): The system should provide a responsive user interface, reliable automation, secure data handling, accurate fare calculation, and consistent ticket-record persistence.

3.3 Data Flow

The basic transaction flow is:

[Passenger / Service Catalog Form] → [Client-Side Validation & Fare Logic] → [ServiceNow Flow Automation] → [Ticket Record & QR Code] → [Payment / Validation / Expiry Tracking]

3.4 Technology Stack

- Platform: ServiceNow
- User Interface: Service Catalog / Service Portal
- Client-side Automation: Catalog Client Scripts and UI Policies
- Workflow Automation: Flow Designer
- Data Storage: Custom ServiceNow tables and ticket records
- Ticket Validation: QR-code-based ticket confirmation
- Scripting: JavaScript within ServiceNow

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The design addresses common metro-ticketing problems through a centralized digital booking form, automated fare calculation, automated ticket generation, QR-code validation, and structured data storage. These features reduce manual intervention and make the booking process easier to track.

4.2 Proposed Solution

The proposed solution introduces a self-service metro ticket booking experience in ServiceNow. Passengers enter their journey details through a Service Catalog item. Client Scripts and UI Policies control field behavior and validation, while Flow Designer automates subsequent ticket-processing activities. The generated ticket contains important booking information such as source, destination, fare, payment status, booking date, expiry, and QR-code information.

4.3 Solution Architecture

The system follows a multi-stage process:

- 1. Passenger opens the Metro Ticket Service Catalog item.
- 2. Passenger enters source, destination, ticket type, and number of tickets.
- 3. Client-side logic validates the input and calculates the fare.
- 4. Flow Designer automates ticket-processing activities.
- 5. Ticket information is stored in ServiceNow.
- 6. QR-code information is generated/validated for the ticket.
- 7. Payment status, booking date, and ticket expiry are maintained.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project implementation was organized into functional development stages so that the ServiceNow application could be developed and validated systematically.

- Stage 1 – Service Catalog Form Setup: Create the metro ticket booking item and required variables.
- Stage 2 – Station Details Mapping: Configure station names, station codes, metro lines, and city information.
- Stage 3 – Fare Logic: Implement Client Script logic for automatic fare calculation.

- Stage 4 – UI Validation: Configure Catalog Client Scripts and UI Policies for dynamic field behavior.
- Stage 5 – Ticket Automation: Configure Flow Designer for automated ticket-processing activities.
- Stage 6 – QR Code Integration: Implement QR-code generation and ticket confirmation/validation.
- Stage 7 – Testing and Validation: Test ticket booking, fare calculation, payment status, ticket generation, QR validation, and data persistence.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing focuses on the responsiveness of the Service Catalog form, execution of Client Scripts, fare-calculation speed, Flow Designer automation, ticket generation, and data persistence. The target is a responsive booking experience with quick processing of normal ticket-booking transactions.

6.2 Functional Test Coverage

- Service Catalog Form and UI Logic – 20 test cases
- Client Scripts and Fare Calculation – 15 test cases
- Flow Designer and Ticket Automation – 15 test cases
- QR Code and Ticket Validation – 10 test cases
- Database Persistence and Security – 10 test cases
- Payment and Ticket Status Validation – 6 test cases
- Total – 76 test cases

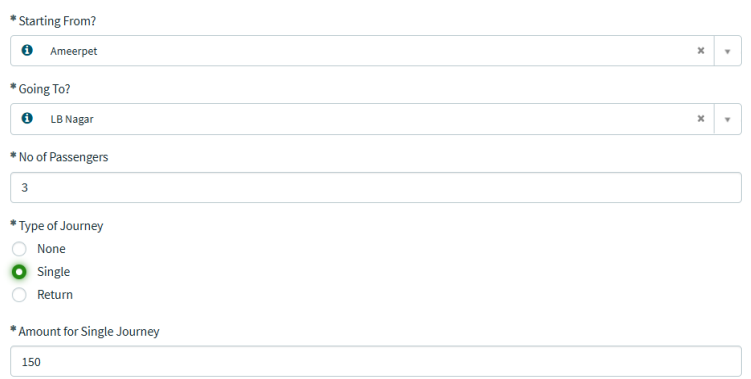
6.3 Test Result

For the prepared project test scenario, 76 out of 76 test cases were recorded as passed, giving a 100% pass rate with zero open failures. The testing covers the main booking, fare, automation, QR-code, payment-status, and data-persistence functions.

7. RESULTS

7.1 Output

- Metro ticket booking form successfully captures the required journey and ticket details.
- Fare is calculated based on the selected journey and number of tickets.



* Starting From?

Ameerpet

* Going To?

LB Nagar

* No of Passengers

3

* Type of Journey

☐ None

☒ Single

☐ Return

* Amount for Single Journey

150

- Ticket records are generated and stored in ServiceNow.

User details	Smart Card Name	Smart Card Number	Recharge Amount	Mode of Payment
Search	Search	Search	Search	Search
6816f79cc0a8016401c5a33be04be441	Mounika	9,876	1,000	BHIM UPI
6816f79cc0a8016401c5a33be04be441	Padma	8,765	850	BHIM UPI

- QR-code-based ticket confirmation/validation is incorporated.



- Payment status, booking date, and ticket expiry are maintained.
- The solution reduces manual ticket-booking effort and provides a structured digital process.

8. ADVANTAGES

Advantages

- Reduced waiting time through digital ticket booking.
- Automated fare calculation reduces manual calculation errors.
- QR-code-based validation provides convenient digital ticket handling.
- Centralized ServiceNow records make ticket information easier to manage.
- Automation reduces repetitive administrative work.
- Payment status and ticket expiry can be tracked systematically.

9. CONCLUSION

The Metro Ticket Generating System in ServiceNow provides a structured digital approach to metro ticket booking and generation. By combining Service Catalog, Client Scripts, UI Policies, Flow Designer, automated fare calculation, ticket records, payment-status tracking, and QR-code-based validation, the system reduces manual effort and improves the passenger ticket-booking experience. The defined testing results demonstrate that the major functional areas of the solution can be validated systematically.

10. FUTURE SCOPE

- Integration with real-time metro station and fare databases.
- Integration with live UPI and other digital payment gateways.
- Mobile-friendly ticket access and notifications.
- Real-time QR-code scanning at metro entry and exit gates.
- Live train and station status integration.
- Analytics dashboards for passenger traffic, revenue, and ticket usage.
- AI-based travel recommendations and demand prediction.
- Integration with smart-card systems for automated recharge and balance tracking.

11. APPENDIX

Key ServiceNow project components used in the Metro Ticket Generating System include the Service Catalog item, metro station details table, Catalog Client Scripts, UI Policies, Flow Designer automation, fare-calculation logic, ticket records, payment status, QR-code functionality, booking date, ticket expiry, and smart-card recharge functionality.

GitHub & Project Demo Link

- Repository: <https://github.com/KesireddyPadma/Metro-Ticket-Generating-System-in-ServiceNow.git>